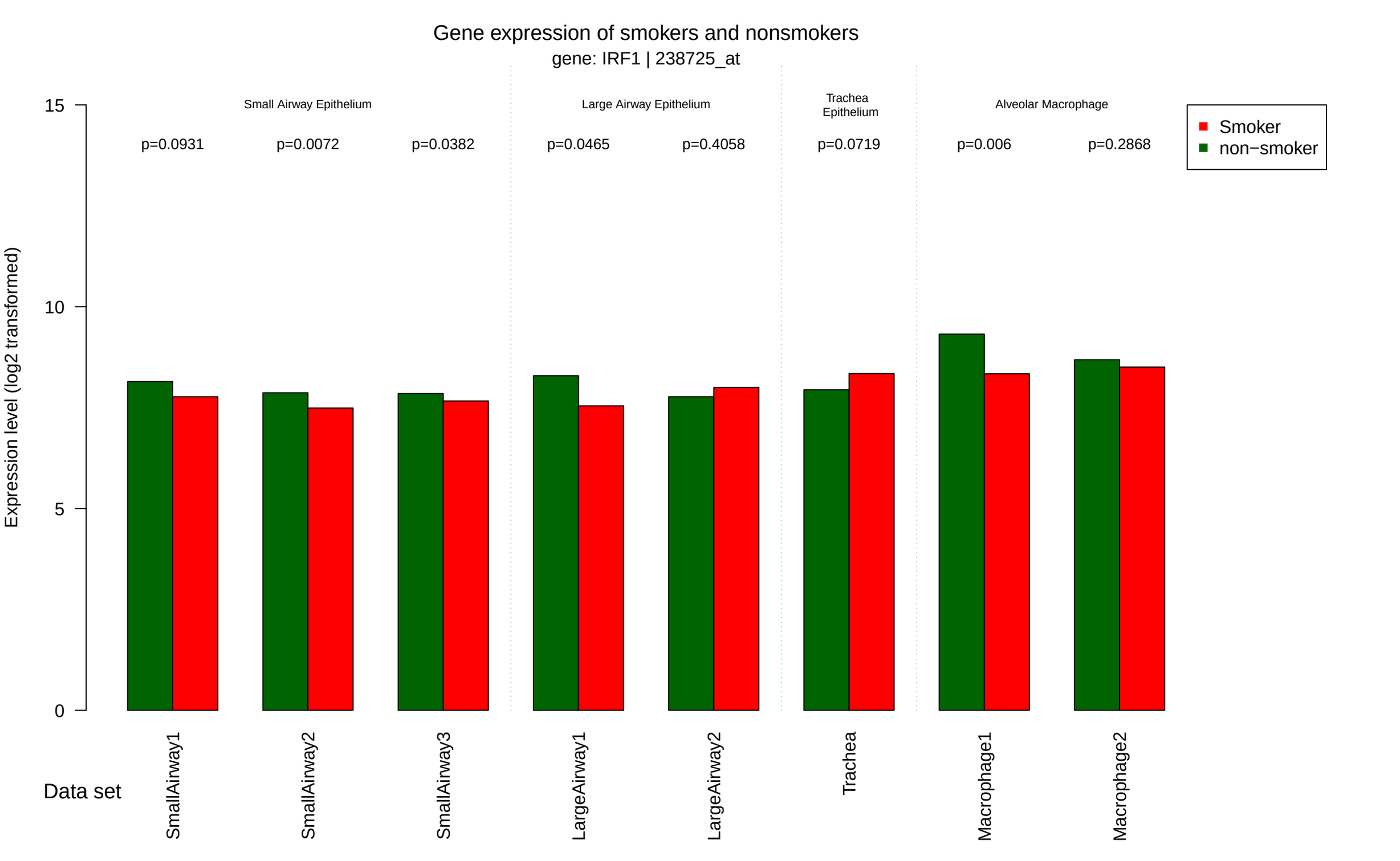
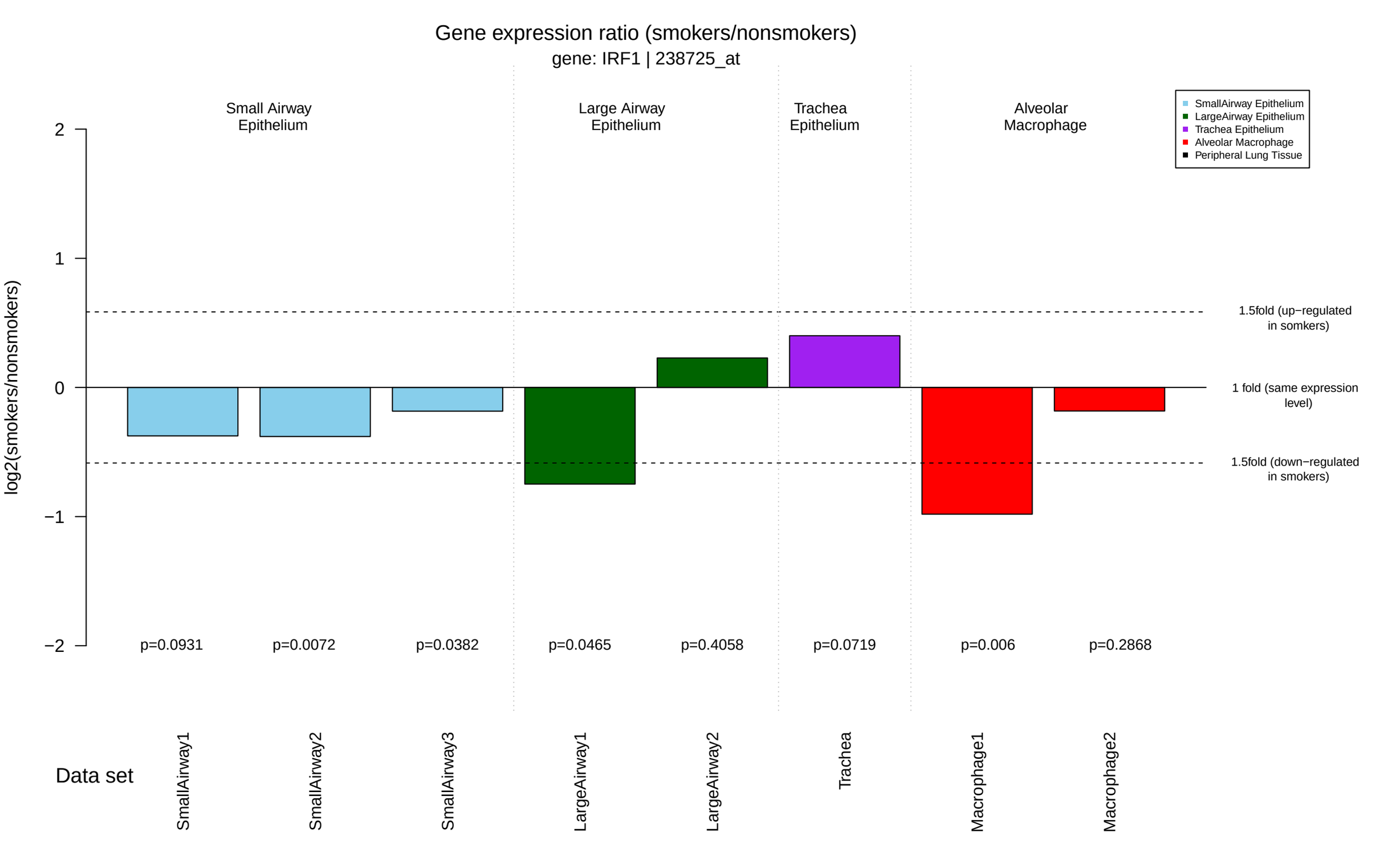
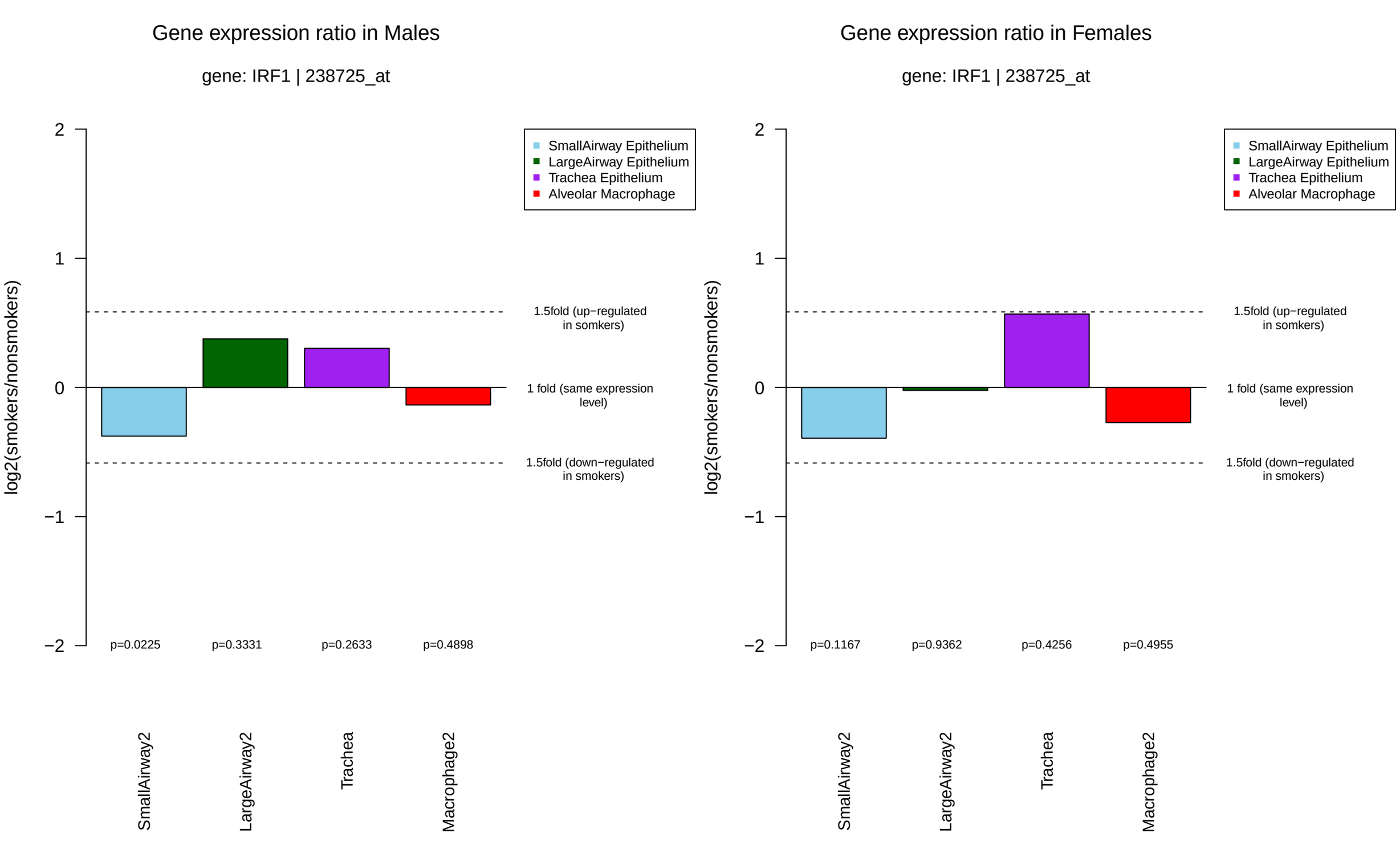
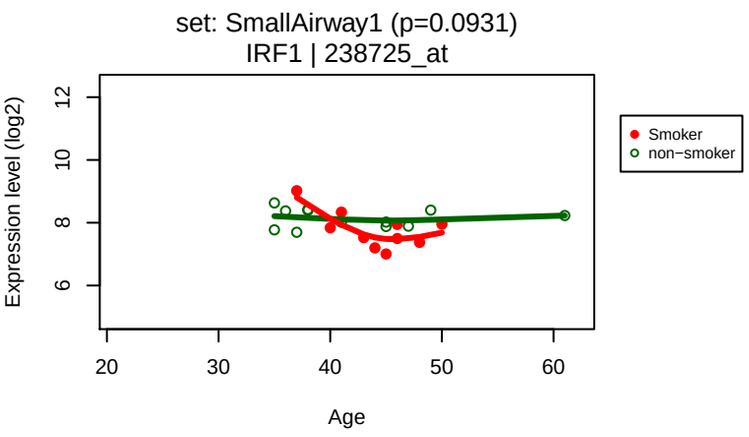
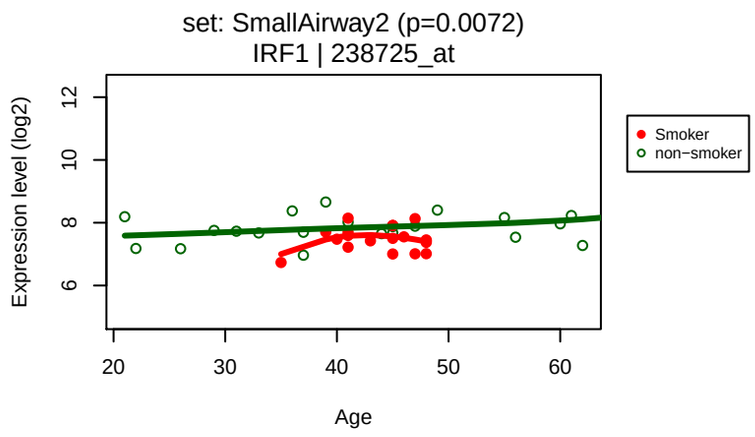
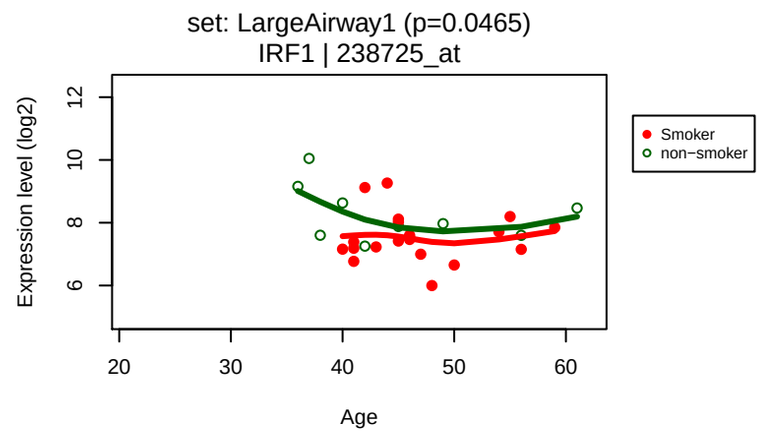

Gene expression of smokers and nonsmokers
gene: IRF1 | 238725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0931
p=0.0072
p=0.0382
p=0.0465
p=0.4058
p=0.0719
p=0.006
p=0.2868
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

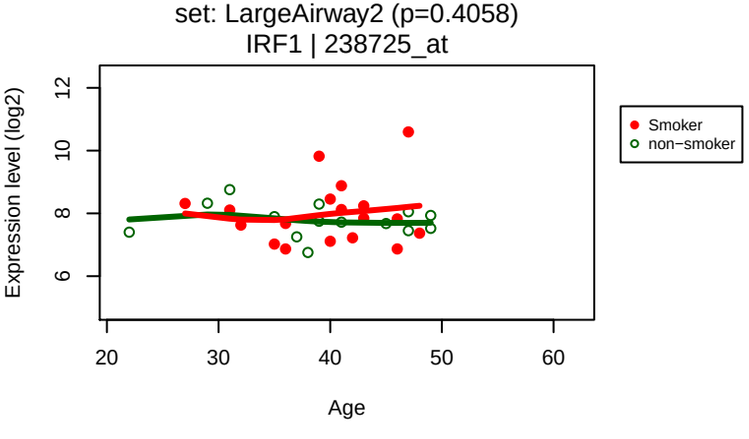

Gene expression ratio (smokers/nonsmokers)
gene: IRF1 | 238725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0931
p=0.0072
p=0.0382
p=0.0465
p=0.4058
p=0.0719
p=0.006
p=0.2868
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

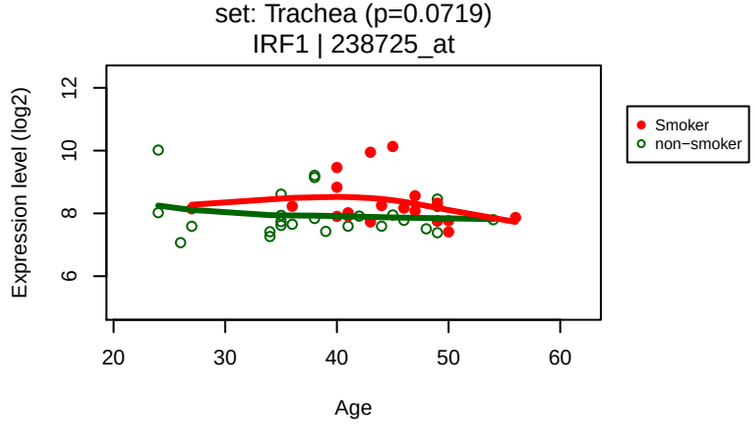

Gene expression ratio in Males
gene: IRF1 | 238725_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0225
p=0.3331
p=0.2633
p=0.4898
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRF1 | 238725_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1167
p=0.9362
p=0.4256
p=0.4955
SmallAirway2
LargeAirway2
Trachea
Macrophage2

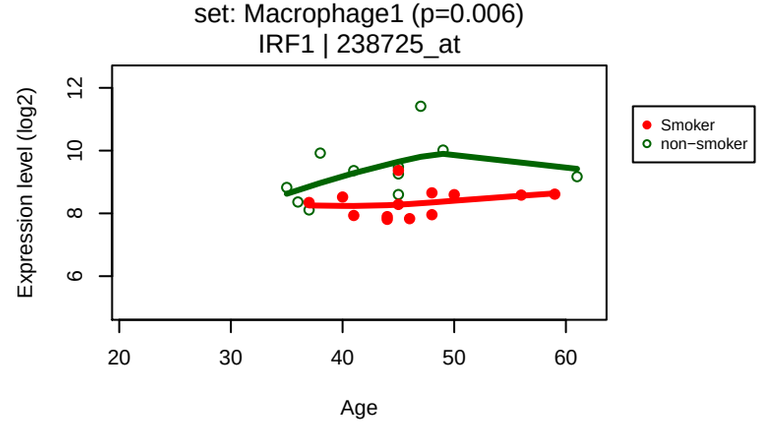

set: SmallAirway1 (p=0.0931)
IRF1 | 238725_at
Expression level (log2)
Age
Smoker
non−smoker

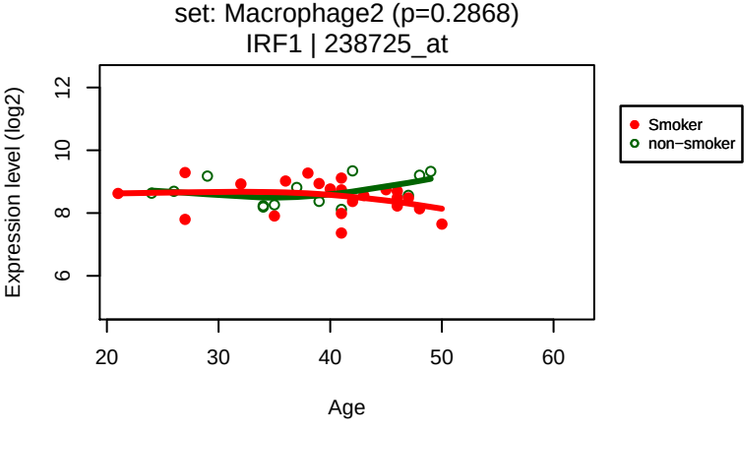

set: SmallAirway2 (p=0.0072)
IRF1 | 238725_at
Expression level (log2)
Age
Smoker
non−smoker

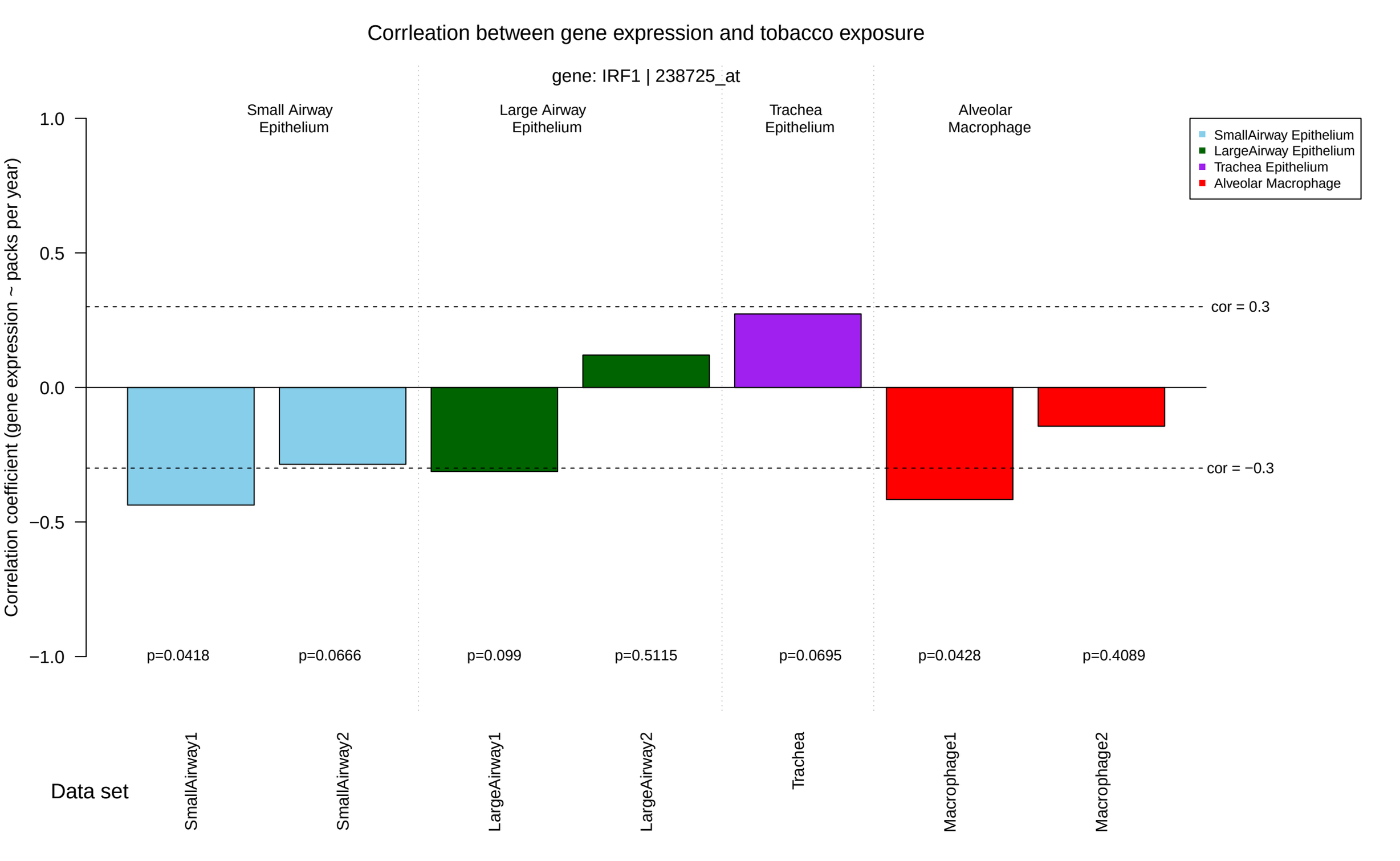

set: LargeAirway1 (p=0.0465)
IRF1 | 238725_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4058)
IRF1 | 238725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0719)
IRF1 | 238725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.006)
IRF1 | 238725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2868)
IRF1 | 238725_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRF1 | 238725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0418
p=0.0666
p=0.099
p=0.5115
p=0.0695
p=0.0428
p=0.4089
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2